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Early Aesthetics Integration in Digital Twin Dashboards in Isolated Agriculture: Data Modelling and Remote Networks

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Abstract—Agricultural enterprises are awakening to the realization of the benefits of digital twins. The rapid integration of dashboard management in remote agricultural locations is driven by the emerging accepted practices involving digital twins. In many cases these amalgamations are limited by agricultural environments that incorporate technology, communication and understand their value in creating optimization, profit, and sustainability. Many environments are insufficiently mature to benefit from the full impact of digital twin modeling, yet farmers, pastoralists and horticulturalists continue to trial and test different technologies because they seek to replicate their rivals and choose to integrate technology as and when individual elements become available, affordable, and applicable. This paper evaluates dashboard processes by case study comparisons. It discusses the decision-making processes that lead to effective deployment of digital twinning and its benefits and posits the importance of aesthetic feature inclusion on dashboards in remote reporting locations.

Keywords—digital twin, aesthetics, AR/VR, dashboard, smart farming, remote networks

I. INTRODUCTION

Agriculture plays a crucial role in global food security and sustainability. As technological advancements continue to reshape various industries, agriculture has not remained untouched. An additional level of understanding over the precision agriculture concept has been developed in the form of information visualization [1], [2], [3], [4]. Such systems can be beneficial as large volumes of data can be easily collected, filtered, or explored for relevant information using interactive imagery [5]. Tasks with significant cognitive efforts can also be solved using visualization tools in combination with tactile, handheld devices and online technology-sharing solutions [6], [7].

The term digital twinning has emerged as an accurate descriptor for the practice of drawing from data in mirrored environments and representing that data in a manner that enables efficient use of resources. The UN Sustainable Development Goal for Zero Hunger (SDG2) has placed global attention on the use of IoTs, Drones and Machinery as part of a world-wide effort to meet the needs of the 2050

expectation of a population of 10 billion people [8]. Through the deployment of digital twin modelling these IoTs, sensors and drones can be mapped, measured, and visualized as a mirror image of an existing environment. The practice of digital twinning is well established in large and sophisticated agricultural establishments, but there is a wide-ranging and persistent response for the world to extend this practice to make use of the inexpensive benefits of technology as part of a broad approach to the achievement of SDG2 [9]. The use of digital twins now enables farming communities the real-time benefit of determining and controlling change, growth, and management issues on any agricultural enterprise. The benefits of such work can be seen in terms of efficiencies and optimization in agricultural decision-making [10].

Projects involving simulated environments, such as, Virtual Reality (AR) and Augmented Reality (AR) based applications have been around since the 1950's. Such virtually simulated environments have become prevalent due to rapid development in the field of computer-based technologies that has reduced costs and delivered easy to use applications [11], [12]. Real time virtual environments provide useful information in the form of virtually generated objects, such as sounds, graphics or texts, through which people can interact to get real experiences through virtual contents. One important example for such scenarios can be the use of AR and VR based applications by the farmers for examining livestock and crops [12], [13], [14]. Research that involve agriculture-based investigations have been improved with the use of digital technologies [12], [14] [15] including visualization technologies. VR and AR based digital visualization technologies within the agriculture-based research contributes towards both betterment of livestock [12], and crop farming, through improved efficiency and reduced supervisory costs [13]. However, such technologies are still in the developmental phase that requires substantial funds and extensive customization to meet high demands [16]. One critical factor that can affect user experience for such applications is aesthetics.

Aesthetics play an important role in terms of how the user experiences products and objects [17], [18], because aesthetics is the representational component of a broader range of sensory experiences such as interest, intrigue, positivity and credibility. Applying appealing and trustworthy aesthetics to certain objects changes their

meaning and overall perception to the user. Aesthetics and functionality both need to be considered by the designers and developers of agricultural dashboard systems that interpret and visualize data from digital twin models [19], [20], [21]. Functionality is important for an individual user's adaptation whereas aesthetics are important for each user's positive perception [22].

This article explores the aesthetics of AR and VR in agriculture, highlighting their impact on productivity, sustainability, and the visual landscape of farming.

II. BENEFITS FROM THE INCLUSION OF AESTHETICS

The use of immersive technologies such as VR in everyday life has increased in the past recent years [23]. AR and VR technologies bring a new dimension to the aesthetics of agriculture by blending digital elements with the natural environment. AR overlays digital information onto the physical world, while VR immerses users into a completely simulated environment [24]. In agriculture, these technologies offer the potential to create visually captivating experiences that enhance understanding and decision-making. This has become possible with the proliferation of modern technological tools such as Unity Game Engine [25], tracking sensors and VR headsets.

A. Application of Aesthetic Weighting

The use of an Aesthetically Weighted Framework can help focus on designing elements that improve user experience and engagement. One such framework, (PCAWF) [26], has been adapted to define the range of aesthetic features and to factor the key areas within an interactive dashboard application. This framework highlights four important elements that can help in defining camera viewpoints and visual representations. Important elements of the framework include the integration of key indices and features, the user state, the type of viewing engagement and the application type. These factors are critical components that define users' engagement and immersion in an interactive digital media application, ranging from AR/VR based applications to dashboard systems that inform users with information from a digital twin model. This paper focuses upon the value of dashboard management systems that engage agricultural users for the purpose of improved and optimized food and livestock outcomes.

B. Data with Aesthetic Features

Dashboard visualization systems along with AR/VR applications provide immersive experience to a user by using the combination of virtual and real worlds as defined in [27]. When the user is surrounded by an immersive and aesthetically enhanced environment, the inclusion helps in elevating awareness, and consequently raises the user's presence and interaction within each application [28]. Engagement is an important factor for such applications along with immersion as acknowledged in [[19], [21], [29], and [30]. Retention and understanding can be enhanced with better immersion that increases user engagement and provides more personalized experience in AR/VR [31]. Many AR/VR applications try to enhance the immersion and engagement in dashboard visualization systems developed for education and training [[28], [32], [33]].

Some dashboard systems draw upon immersive technologies like AR/VR. They are being developed for various types of education, training, and edutainment

applications. For example, VR has been used for training medical students and surgeons for performing biopsies [34], in VR gyms for physical fitness [35], and in immersive simulation software for teaching youths to operate tractors [36]. There are various benefits from using AR/VR for educational and training purposes. These include improved accessibility (mobility, visual, auditory) [32], they provide more immersive and interactive learning experiences [37], and they are safe for training young students [36].

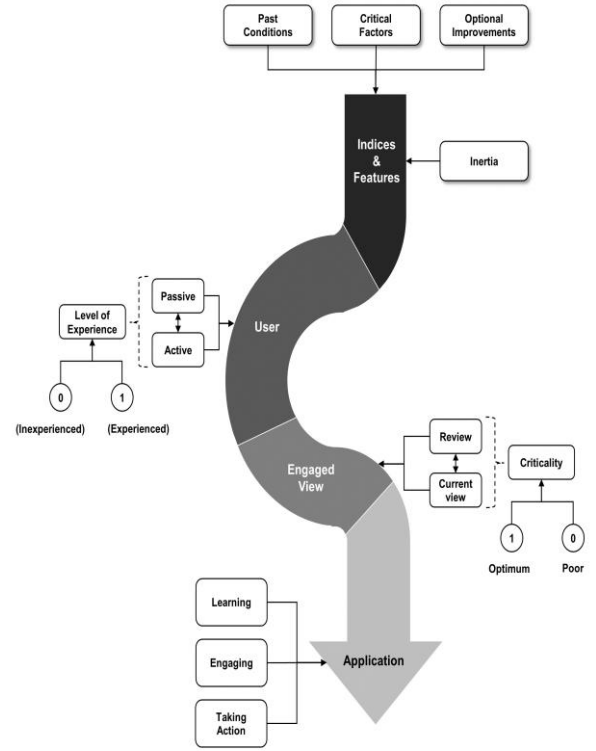


Fig. 1. Data with Aesthetic features for Dashboard Management (DADM), adapted from H. Maqbool, 2023.

Aesthetic visualizations are critical inclusions in sustainable decision making when using remote networks and relying upon low-bandwidth numerical data. Substantial advancements have been observed in AR/VR technology in recent years. This technology has the potential to transform different areas [37] including dashboard systems when used for decision-making purposes. It is easily extendable into the application of AR/VR in Agriculture [[38], [36], [39], [40]]. Use of such technologies will help quantify and monitor agriculture in order to reduce the overall cost, and improve yields for the farming sector [38]. Aesthetics can play a very important role in this regard as it will help in improving the overall user experience with increased immersion and engagement. This can be achieved if there are frameworks like PCAWF and DADM that help a developer to focus on aesthetic elements of AR/VR technological applications.

C. An Aesthetic Framework for Digital Twins

Many AR/VR applications have been developed in different fields. The inclusion of aesthetics in the deployment of digital twin modelling plays an important role in improving veterinary sciences. The concept of digital twinning is increasingly entering livestock farming strategies.

This allows for development in technology, and encourages expansion in the number of animals on farms required to meet needs of human population and the ongoing trend towards urbanization [40].

The integration of digital twin modelling results in the creation of new educational resources based on accurate data referring to animals [40] and crops [41]. Data on various key livestock performance parameters, such as the health, weight, productivity, and environmental conditioning of animals is collected. Similarly, crop health and productivity can also be observed and recorded. At the same time, AR and VR based technologies and learning tools are emerging to help novice breeders and small-scale farmers gain experience in safe environments. By incorporating these educational and training tools before working in the real world there is a substantial reduction in risks, errors, and expenditure.

Digital twin approaches can benefit farmers by providing significant practical information and data. This is exemplified in the managed usage of chemicals, fertilizers, and irrigation, in combination with an increased appreciation of improvements to seeds, environmental protection, market demands, and business conditions [42]. In animal farming the same technology can be used for foreseeing and preventing harmful behavior of animals, such as tail biting, feather pecking, and horn strikes. Farmers can learn the responses of chicken and pigs to the environment, consequently they can take actions accordingly for improving the welfare of their livestock [43]. Such an approach is highly dependent upon the farmers' adoption of the visual features and numerical data, making sense of the way in which they demonstrate the opportunities and the need for specific data-driven actions on each farm. The inclusion of aesthetic qualities becomes critical to persuading individual farmers to rely on digital data and to use the information to make informed decisions that are founded upon that data.

The development of digital twin AR/VR based applications requires deeper understanding in order to apply aesthetics. One aesthetic element that can be focused on such applications is the creation/visualization of real-time data representing the exact location for the collected data. This has a "flow-on" effect to industry by providing faster learning, richer knowledge, and safer environments. It is also useful in the prevention of harmful or unnecessary experiments on livestock.

D. Remote data dashboard analysis for Digital Twins

One of the strongest emerging trends in the early adoption to digital twin modelling is the notable difference between research-based interpretation and "in-the-field" dashboard interpretation. When early versions of digital twin systems began to demonstrate the benefits of modelling agricultural information from sensor data, it was carried out by research and data scientists who were able to deploy precision agriculture initiatives, and who had a clear scientific appreciation of the benefits of improved soil science [44]. A useful comparison to agricultural digital twins can be found in manufacturing production processes. Here there is a strong alignment because both manufacturing and agricultural production can benefit significantly from dashboards that provide in-depth data analysis and decision making. In

addition, both remain exposed to the potential vulnerability of standstills in production [45].

In contrast, the emerging trends demonstrate the actions of farmers and their interpretations based on dashboard information [46], [47]. As digital twin systems have evolved to offer *first to market* dashboard applications, they have either provided sterile numerical updates or they have enticed farmers with generic and heavily summarized analysis which is dependent upon the payment of an additional fee. Many of these analyses provide broad information with the upsell of a collective summary of data. This data usually includes a farmer's data but also summarizes the information with other nearby or similar enterprises. Some practitioners suggest that these dashboard offerings are too generalized to provide specific assistance to individual farmers [48].

III. AR/VR CASE STUDY IN AGRICULTURE

Given below are several case studies that inform the principal of aesthetic visualization and the importance of digital twin-driven decisions. This research emphasizes the need for aesthetically composed visualizations to improve the user experience by proposing the adapted framework that is based on the PCAWF framework stated in [26] for AR/VR based applications in agriculture.

A. Digital Twin Dashboard exemplars

An important consideration is the way in which dashboard information is consumed. This section outlines the needs-basis for dashboard integration based on assumptions drawn from current dashboard examples and best practice.

B. Dynamic Water Consumption

In Australia the conditions favor a strong and watchful approach to water management. In the summer months water becomes a precious commodity and requires a vigilant approach to its usage, depletion, and protection.

Some dashboard management platforms report the numerical quantities only, citing "x" number of kiloliters in a single tank, or "y" number of liters in a dam. These representations are a useful start but are insufficient to offer the required information, or to engage the dashboard consumer, towards the needs of the farm or agricultural output. The market is looking for aesthetically inclusive data that is richer and broader in application, and which allows for predictive AI modelling.

There is a need for an aesthetic dynamic. This ensures that the consumer sees the daily / hourly figures – but more importantly is diverted towards the trends that will suggest extrapolative action and projection-based planning [49]. The ability to see information beyond "quantity only" data is required to engage farmers to develop meaningful strategies for optimized water management. The inclusion and creation of water control tactics needs planned intervention in a timely manner that looks beyond water loss and instead becomes inclusive of livestock movement, combined water source management strategies, and holistic data-driven approaches (e.g., Dams, Water Tanks, Rivers, and Artesian sources).

C. Dashboards in Agriculture

One case study includes the dashboard, (named Citrus Yield), which was deployed to analyze individual fruit after collection as they are moved through a conveyor belt. This prototype incorporated a visualization system that was used

to inspect the quality of produce artificially while they were harvested [44]. This dashboard consisted of various inputs in the form of quantitative information, such as crop and harvesting information.

The advantage of using Citrus Yield as a dashboard was that it automatically generated yield and quality maps for fruits. The maps indicated various factors that could have affected the fruit quality such as irrigation, nutrients, and soil quality. Another advantage of using such a dashboard is that the information can easily be processed and obtained without any computer skills [50].

The main disadvantage of this system was that this implementation was only applied across citrus fruit. This provided a limitation to other crops such as wheat and barley. This system demonstrated how certain camera angles were used to analyze fruit quality. This case study demonstrated the benefits of looking beyond numerical data and integrating combinational approaches. Remote access to crop data was greatly improved in cases where the visual appreciation of the data went beyond numbers and allowed the dashboard reader to visually make sense of the numerical data.

D. Precision Farming using AR/VR

Precision Farming is responsible for generating data analysis that involves interpretation using the data and making decisions to enhance crop production [51]. Applications based on AR/VR were used for acquiring data systematically. This data could be used to interpret the important information from critical results on the fields.

The research explored in [51], [52], [53] reviewed the case of precision farming. In [51] the AR application was used to analyze the diseases of the plants, whereas in [52] the research focused on the effect of camera viewpoints and orientation of fruit for sweet pepper maturity rather than disease.

The study in [51] for plant disease detection incorporated an AR approach using a cloud-based machine learning system. A head-mounted display was used by the farmers for observation of fields. Specific data and information related to plants was viewed virtually through the visual data stored in the cloud. In short, plant disease was viewable by farmer workers using an AR system. This enabled farmers with little or no knowledge in plant disease diagnostics to view and recognize areas of concern and to make decisions. This study used machine learning for the classification of plant foliage. This example demonstrated the possibility of bottlenecks since a large amount of data was required to train systems for diagnosing different disease types.

The study in [52] focused on the choice of the best viewpoint by selectively reducing the number of chosen viewpoints using a performance and cost criteria for the deployment of harvesting robots. This study was divided into three parts in a controlled laboratory environment. In the first step the analysis of viewpoints acquired for sweet pepper images was carried out in a photocell. The second step consisted of analyzing the performance of a robotic arm for sweet peppers attached to plants, and in the last step a classification algorithm was tested for different orientations. This study demonstrated that the choice of viewpoints for a maturity test is critical for crops that have non-uniform maturity patterns, such as sweet peppers. To assess the

maturity of crops accurately the estimation of orientation is very important for the choice of a viewpoint used for classification. If orientation is known in advance, then the framework can be used to estimate the correct viewpoint.

Both these analyzed systems used AR/VR-based technologies to assist farmers in precision farming. Both systems required certain type of viewpoints for plants to do the analysis.

E. Crop Monitoring using AR/VR

With the increase in use of Computer-Automated Controlled Environment Agriculture (CEA) [[48], [49], [50]] improvements have occurred in green houses. This application has maximized production using daylight and seasonal management. As there is an increase in use of smart technologies for greenhouses, there is a need for training that specializes in management and use of CEA systems. Due to the potential loss of yield because of mismanagement, students are sometimes restricted from working with CEA technologies and their associated parameters. This has resulted in the lack of real-world interactions for students working on such systems, and has also reduced its effective use for profitable causes [48].

One solution to overcome this problem can be the use of virtual greenhouses for education and training. One such greenhouse has been proposed in [48] where a VR has been used for educating students. In such scenarios aesthetics can play an important role for defining the level of engagement and realism to improve users' learning.

The study in [51] uses AR based educational software whose purpose is the development of agriculture based techniques in reference to aquaponics. This application emphasized the fact that such applications can potentially motivate users to learn agricultural techniques. Subsequently this also suggests that it is suitable to encourage organic agriculture strategies in aquaponics.

This problem can be overcome by the use of the DADM framework [26] where camera angles for immersive applications allow for a richer engagement with the numerical data. The resulting benefits from an aesthetically inclusive dashboard are in the form of improved data visualization.

IV. APPLYING AESTHETICS USING A FRAMEWORK

Aesthetic elements that AR/VR technologies introduce to agricultural environments could include visual immersion in case of educational tools, user interface design, and integration of real and virtual world. While the aesthetic potential of AR/VR in agriculture is promising, challenges such as cost, accessibility and technological integration remain unresolved. Ensuring that these technologies are user-friendly and adaptable to varying farm environments is essential to maximize their impact.

A. Using the DADM Framework to increase immersion with dashboard and digital twin management

Enhancing aesthetics in the context of Augmented Reality (AR) and Virtual Reality (VR) for digital twin applications is of paramount importance in creating experiences that are not only visually appealing but also captivating and immersive. The DADM framework [26] can play a pivotal role in

achieving these goals, particularly in the context of education-based applications.

In the realm of digital twin applications, where virtual representations mirror real-world entities, creating an intuitive and engaging user experience is essential. This is particularly true for educational purposes, where users often seek to comprehend complex concepts, visualize scenarios, and grasp spatial relationships. The DADM framework can contribute significantly to achieving these objectives by offering a structured approach to designing camera viewpoints within virtual 3D environments.

The DADM framework encourages thought-provoking visualization and careful structure of scenes. For educational digital twin applications, this means selecting camera viewpoints that provide clear and comprehensive perspectives, allowing users to understand spatial relationships and context. The framework's recommendations align with user-centered design principles, focusing on delivering experiences that cater to learners' needs, preferences, and cognitive processes.

V. SUMMARY OF AESTHETIC INCLUSION BENEFITS

The inclusion of aesthetics in various contexts, such as design, technology, and user experience, offers a range of benefits that contribute to enhanced visual appeal, user engagement, and overall effectiveness. The benefits incorporate enhanced visual curiosity, user engagement, positive user experience, clear communication, easier learning, efficient navigation, accessibility, and stress reduction. They are important outcomes from the application of relevant aesthetics for the visualization of dashboard applications in agriculture [54], [55].

Incorporating aesthetics into various aspects of design, technology, and user experience, can have a profound impact on how people interact with and perceive products, services, and information. The benefits of aesthetic inclusion extend beyond superficial attraction, enhancing engagement, communication, and overall quality of experiences [56], [57].

VI. CONCLUSION

The aesthetics of AR/VR are reshaping the agricultural landscape, enriching the visual experience of farming practices. Remote sensing and digital twinning have grown exponentially but require dashboards that take the information beyond numerical limitations and allow users (especially those in farming) to make decisions that are more richly integrated with real-time, on-demand, data. As technology continues to evolve, the balanced fusion of digital and natural elements in agriculture will likely inspire new generations of farmers, consumers, and stakeholders. Beyond their utilitarian and functional benefits, these technologies open the realms of aesthetic possibilities that optimize the natural positive values and qualities of agriculture whilst driving and influencing efficiencies and sustainable practices.

Future work will focus on specific areas of agricultural challenge and global concern. As IoTs become the ubiquitous provider of low-cost streams of information, the nexus between data-driven models and visual decision-making requires the ongoing integration of the aesthetic values of

numerical data and information. The benefits of aesthetically driven climate-change data will allow for greater engagement and focus on areas of critical development and urgency. Food security will benefit from an intensified focus upon sustainable optimization techniques on food production and output.

The future research vectors for digital twin dashboards will apply the DADM model into detailed and specialized areas where dashboard management by farmers requires uplift, technology acceptance, and application.

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